Shot Estimation

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Introduction

- For current NISQ-era computer shot estimation remains a difficult problem.
- There has been work in the past analysing the theoretical convergence of specific circuits.[1], and statistical properties of qubits[2] to get better estimates. The general approach however has been trial-and-error[3]
- Here we will run a statistical analysis on various forms of noise, to then be able to estimate how many shots are required to obtain a result within known error bounds.

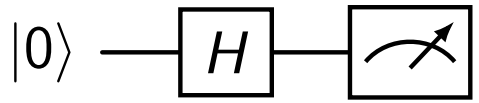
[1] Peruzzo, McClean, et al. 2014; Sweke, Wilde, et al. 2020.
[2] Burnett et al. 2019.
[3] Sung, Yao, et al. 2020; Brown, Goktas, et al. 2020.

Plan

- Establish a baseline for a single-qubit circuit for with one H gate and measurement.
- Add each source of noise individually to the baseline and plot the results
- Find patterns in variations from basline for each source of noise and make a prediction mechanism
- We expect to end with a formula for accurate shot estimation within known error bounds

Introduction

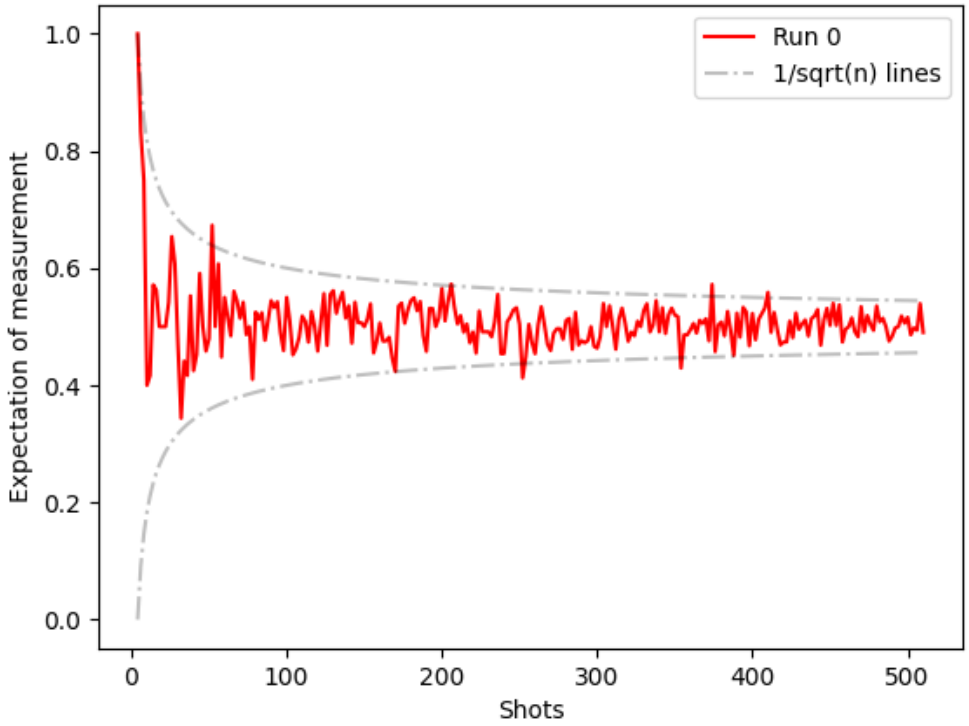
- Consider a single-qubit circuit with one H gate and measurement as our baseline.



- Over some n measurements we will obtain a distribution with probabilities of $0, 1$.
- We find the expectation of the measurement X as $\mu = \mathrm{E}[X] = \sum_{i=1}^{\infty} x_i p_i$ and plot μ vs n

Baseline

- Number of shots (2 to 512 shots)



- Width at any n decreases as the number of shots (n) increases, as $1/\sqrt{n}$.
- Compare this plot to noise models of isolated sources of noise.

Common sources of Noise

- **SPAM Noise** - State Preparation And Measurement errors arise either during the preparation of the qubit, or during measurement.
- **Relaxation Noise** - Qubits will decay to the ground state over time i.e $|1\rangle \rightarrow |0\rangle$.
- **Dephasing Noise** - Qubits will lose phase information overtime and go to an equal superposition of $+\rangle$ and $|-\rangle$.
- **Gate Noise** - Errors in the application of gates single and multi-qubit gates
- **Crosstalk Noise** - Errors in the application of gates due to qubits being too close to each other.
- **Correlated Errors** - Other sources of noise that are not independent of each other.

Crosstalk and Correlated errors have no widely accepted models[4]

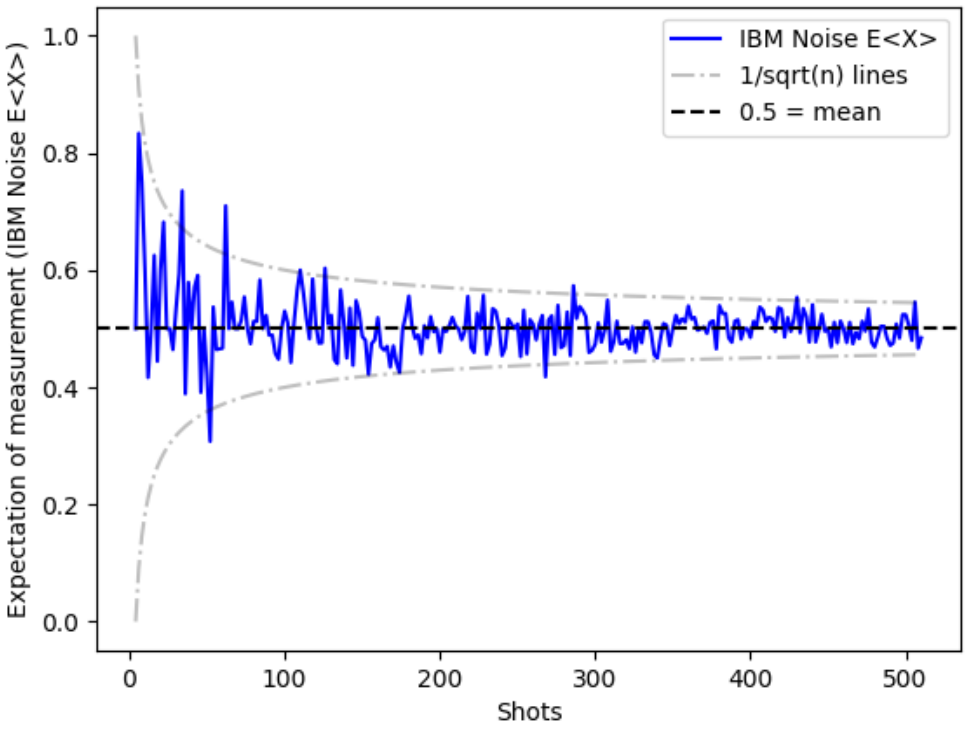
[4] Bravyi et al. 2018; Greenbaum and Dutton 2017.

SPAM Noise

- State Preparation And Measurement (SPAM) errors arise either during the preparation of the qubit, or during measurement.
- Noise simulators only allow us to simulate readout errors, not preparation errors.
  - We will only consider the measurement error in this case.
- Readout errors are made up of two components,
  - probability of a 0 read as a 1 i.e $p_{0\to 1}$,
  - probability of a 1 read as a 0 i.e $p_{1\to 0}$.

SPAM Noise

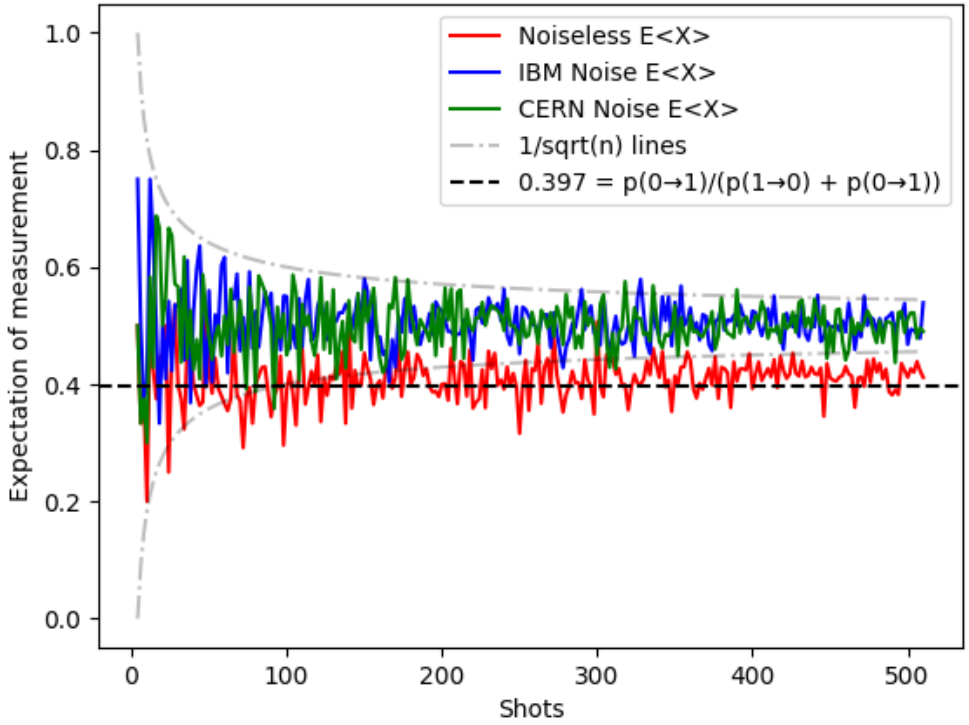
Plot for $p_{0\to1} = p_{1\to0} = 0.33$ on IBM's noise simulator with same baseline circuit.



The following slide has all plots on noise-free, IBM's noise, and CERN's noise simulator.

SPAM Noise

- Increase in width is not visually obvious, so we add asymmetric noise to IBM's simulator to visualise it's effects. for $p_{1\to 0} = 0.5$ and $p_{0\to 1} = 0.33$.



- New mean $\mu' = p_{0\to 1}/(p_{1\to 0} + p_{0\to 1})$ is shifted down when $p_{1\to 0} \neq p_{0\to 1}$.

Difference from mechanism so far

- Instead of taking mean/stdv of n shots, we take mean/stdv of windows of w shots
- New means for N windows of size w form new distribution $\bar{X}$.
- Ex, if we have results of 1024 shots, we would take say $N = 4$ windows of say width $w = 16$ shots each.

$$\text{Windows} = \{[0, 16], [16, 32], [32, 48], [48, 64]\}$$

- To get means of windows, $\mu_{w,1}, \mu_{w,2}, \mu_{w,3}, \mu_{w,4}$.
- New random variable $\bar{X} = \mu_{w,i}$ is now a normal distribution via central limit theorem as $w \to \infty$.

Difference from mechanism so far

- CLT Example: if $X \sim \text{Poisson}(\lambda)$, then $\frac{\bar{X}-\lambda}{\sqrt{\lambda}} \to N(0,1)$ as $\lambda \to \infty$
- So in our case $X \sim \text{Binomial}(n,p)$, then $\frac{\bar{X}-np}{\sqrt{np(1-p)}} \to N(0,1)$ as $n \to \infty$
- Our distribution was not originally normal, we created a new distribution $X \to \bar{X}$ to make it normal and therefore got fixed $-\frac{1}{2}$ slope on a log(RSD) vs log(w) plot as we will see.

Applying CLT Mechanism

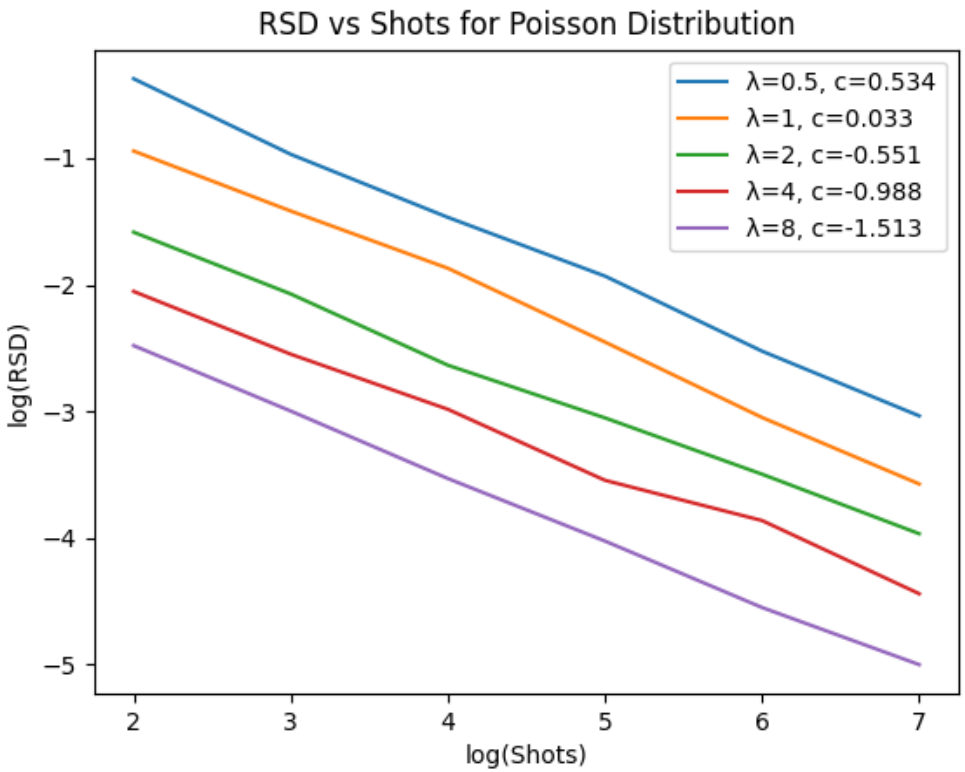
If we were to apply the CLT mechanism to the SPAM noise, and run the following experiment, we do the following procedure.

1. Run the circuit for 2^{15} shots (with $p_{0\to1} = p_{1\to0} = 0.33$)
2. Divide the results into first $N = 256$, windows of $w = 2^2$ shots
3. Calculate the mean of each window, to get a new distribution of 256 values which is normal
4. Calculate the mean, stdv of this new distribution
5. Repeat process for windows of size $w = \{2^3, 2^4, 2^5 \ldots 2^{11}\}$
6. Plot the log(RSD) vs log(w) for each window size [5]
7. We should see a fixed $-\frac{1}{2}$ slope for each window size with varying intercepts

[5] Note here we're plotting against w and not n. Number of windows taken is fixed at $N = 256$

Applying CLT Mechanism (Poisson)

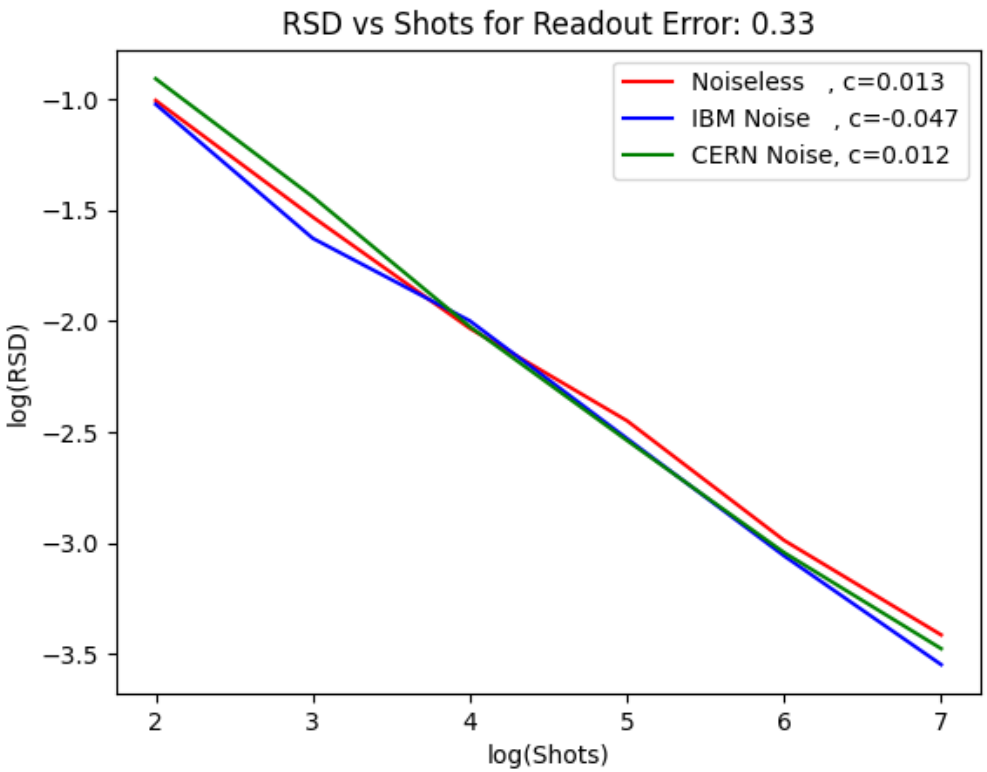
- CLT mechanism on Poisson distribution as an example
- $\lambda \in \{0.5, 1, 2, 4\}$, $n = 2^{15}$ shots



- As λ doubles, each time the line moves down but at a lower rate
- The slope of the line is fixed at $-\frac{1}{2}$ for all λ

Applying CLT Mechanism

- Set $N = 2^8$ windows to get as straight a line as possible to accurately measure the intercept c
- 2^8 implies, for 2^{15} shots, the maximum window size is $w = 2^7$ shots giving us 6 data points $w = \{2^2, \ldots, 2^7\}$



- All intercepts were calculated as $c = \text{mean}(y - mx) \because m = -\frac{1}{2}$

Expected Result

$$
\begin{aligned}
c &= y - mx && (1)\\
&= \log(\sigma/\mu) + \frac{1}{2}\log(w) && (2)\\
&= \log(\sqrt{wp(1-p)}/wp) + \frac{1}{2}\log(w) && (3)\\
&= \frac{1}{2}\log(\frac{1-p}{p}) = -\frac{1}{2}\log(\frac{p_0}{p_1}) && (4)
\end{aligned}
$$

- Since $p \in (0,1)$, $c \in (-\infty, 0)$ as p increases, the line shifts down.
- Therefore we expect $c_2 - c_1 = -\frac{1}{2}\log(\frac{1-p_1}{p_1}\frac{p_2}{1-p_2})$
- Or $c_2 - c_1 = -\frac{1}{2}\log(\frac{1+p_{0\to1}-p_{1\to0}}{1+p_{1\to0}-p_{0\to1}})$
- All symmetric SPAM noise models should have $c = 0$.

Applying CLT Mechanism

- Readout Error $= (p_{0\to1} + p_{1\to0})/2$
- Readout Bias [6] $= p_{1\to0} - p_{0\to1}$ (Coming Down - Going Up)
- We can then use expected c values as $-\frac{1}{2}\log(1 - \text{Readout Bias}/1 + \text{Readout Bias})$

Qubit	T_1	Readout Error	$p_{0\to1}$	$p_{1\to0}$	Readout Bias
61	232	0.0082	0.0099	0.0066	0.0033
129	232	0.0166	0.0186	0.0146	0.0040
5	232	0.0389	0.0350	0.0428	-0.0078

Qubit Parameters from IBM Torino's calibration

[6] Non-standard terminology

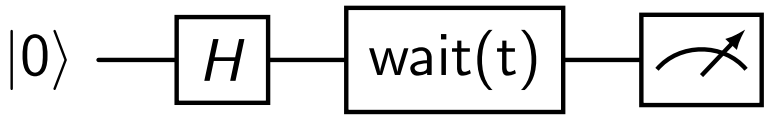
SPAM Results

Qubit	Expected c	Actual c
61	-0.0095	-0.125
129	-0.0115	-0.012
5	0.0225	0.033

- We're able to predict values of c within 0.01 of the actual value from just the calibration data
- Something went wrong with qubit 61.
- The possible reasons could be
  - Since it ran after two days in queue, the calibration data was outdated.
  - Single qubit gate error (Application of H), may have contaminated the results

T_1

- T_1 makes a qubit state decay over time as $P(|1\rangle) = e^{-t/T_1} p_1$ where t is the time since preparation.
- We will add Id^k before measurement for $Id^k = \{10^1, 10^2, 10^3\}$ of T_1.
- We will also vary for various values, subject to system constraints.
- Circuit used will be



Expected Result (Derivation)

- With $p_{1\to1} + p_{1\to0} = 1$, $p_{0\to0} + p_{0\to1} = 1$ and $\epsilon = 1 - e^{-t/T_1}$,
- And $p_0' = p_0 + p_1\epsilon$ and $p_1' = p_1 - p_1\epsilon$
- we can put these into the modified SPAM formula as
- $p_0^F = p_0' p_{0\to0} + p_1' p_{1\to0}$ and $p_1^F = p_0' p_{0\to1} + p_1' p_{1\to1}$
- Therefore

$$c = -\frac{1}{2}\log\left(\frac{b(1-p_{0\to1}) + p_{1\to0}}{b(p_{0\to1}) + (1-p_{1\to0})}\right), \text{ where } b = \frac{1+\epsilon_1}{1-\epsilon_1} \quad (5)$$

[7]

We can see this formula very cleanly regresses to just SPAM noise when there's no noise.

[7] More generally $b = \frac{P(0)}{P(1)}$

Results

- Instead of listing all results we'll map it onto the processor architecture.
- We color code all results as $\Delta_\epsilon c = |c_{\text{pred}} - c_{\text{real}}|$

$\Delta_\epsilon c$	σ Ratio	Color	Meaning
$0.0 - 0.05$	$1 - 1.03$	green	Correct Prediction
$0.05 - 0.1$	$1.03 - 1.07$	yellow	Approximate
$0.1 - 0.5$	$1.07 - 1.41$	orange	Other noise
$0.5 - 1.0$	$1.41 - 2$	red	Bad Qubit, Other noise
> 1.0	> 2	black	Bad Qubit, Wrong Model

- Experiments are H gate followed by n of H, Id, X gates where $n \in \{10^1, 10^2, 10^3\}$.

Conclusions before the Diagram

- We're within 0.03 of the true result for the best qubit
- Errors: Single Connected Qubits $<$ Double Connected $<$ Triple Connected indicating crosstalk
- Errors: Id results $<$ X results $\ll$ H results indicating single qubit gate noise in X, H
- Bad qubits and bad connections show up reliably as errors in even neighboring qubits
- H gate is fundamentally different when being applied

Some conclusions not visible in the diagram are

- We overestimated the error for X experiments for almost every case, indicating that IBM is doing error correction at the hardware/gate level.

Results

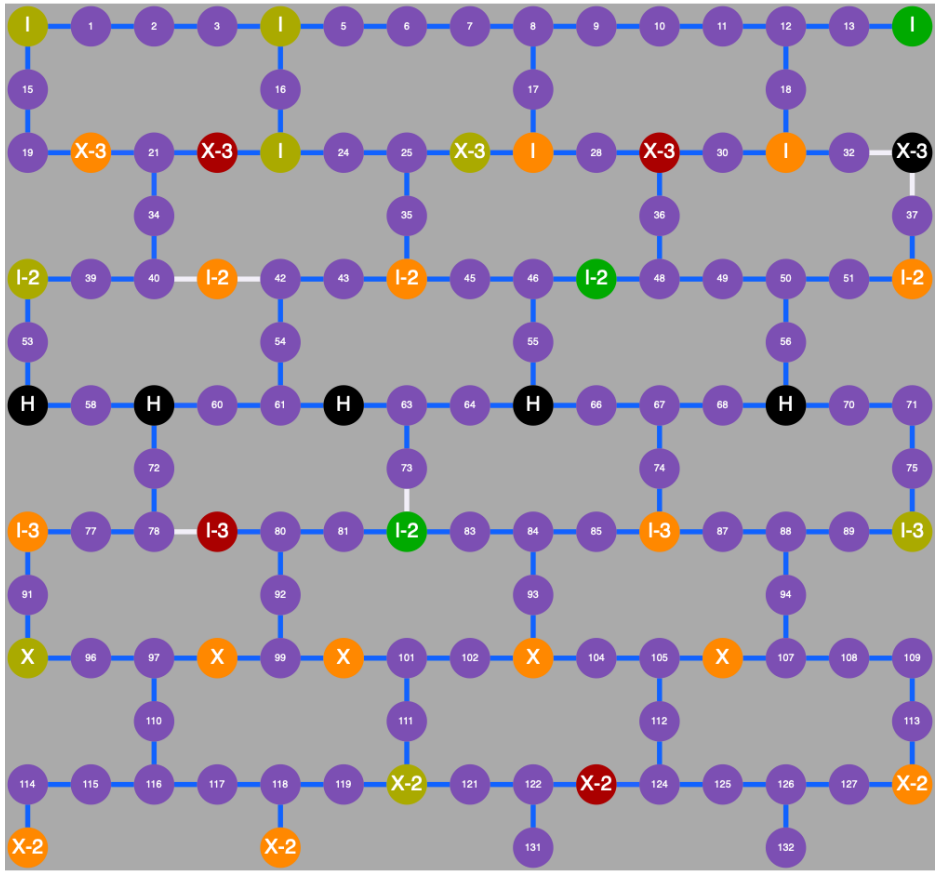


Figure: T1 results

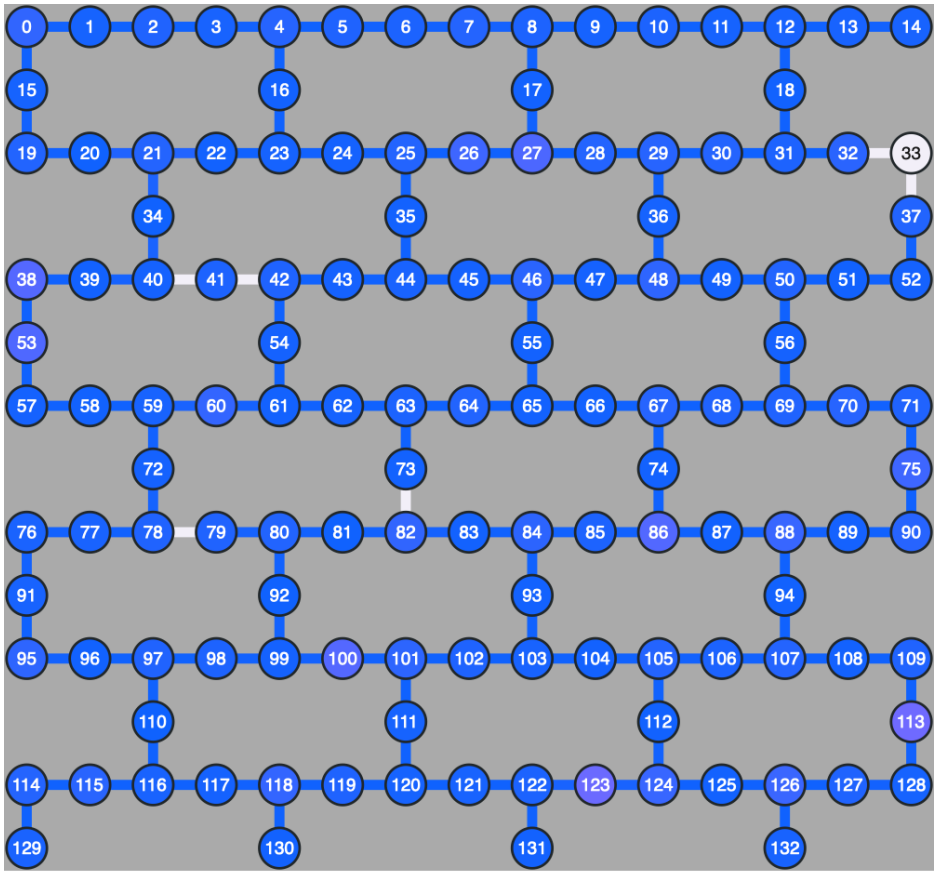


Figure: Torino System Map

cbit	qbit	c	gate	t1	t1 + spam	match
$t1_2$	69	2.021	10h	209	0.0057	(under) 2.0153
$t1_3$	65	2.298	10h	227	0.0194	(under) 2.2786
$t1_4$	62	2.869	10h	128	0.0608	(under) 2.8082
$t1_5$	59	2.176	10h	107	0.0466	(under) 2.1294
$t1_6$	57	2.221	10h	177	-0.0007	(under) 2.2217
$t1_7$	33	0.03	1000x	45	2.1572	(OVER) 2.1272
$t1_8$	29	-0.129	1000x	144	0.5077	(OVER) 0.6367
$t1_9$	26	0.139	1000x	219	0.2355	(OVER) 0.0965
$t1_{10}$	22	0.127	1000x	70	1.1262	(OVER) 0.9992
$t1_{11}$	20	0.307	1000x	172	0.5103	(OVER) 0.2033
$t1_{12}$	128	-0.227	100x	184	0.0158	(OVER) 0.2428
$t1_{13}$	123	-0.036	100x	91	0.8008	(OVER) 0.8368
$t1_{14}$	120	0.024	100x	111	0.0759	(OVER) 0.0519
$t1_{15}$	130	0.329	100x	249	0.0468	(under) 0.2822
$t1_{16}$	129	0.283	100x	225	0.061	(under) 0.222

cbit	qbit	c	gate	t1	t1+spam	match
$t1_{18}$	103	-0.235	10x	244	0.0456	(OVER) 0.2806
$t1_{23}$	86	0.319	1000id	164	0.6838	(OVER) 0.3648
$t1_{24}$	82	0.647	1000id	134	0.651	(OVER) 0.004
$t1_{29}$	44	-0.288	100id	248	0.0371	(OVER) 0.3251
$t1_{32}$	31	0.319	10id	179	-0.0194	(under) 0.3384
$t1_{33}$	27	-0.041	10id	150	0.117	(OVER) 0.158
$t1_{34}$	23	-0.033	10id	159	0.0363	(OVER) 0.0693
$t1_{36}$	4	0.161	10id	153	0.0795	(under) 0.0815
$t1_{17}$	106	0.281	10x	135	0.0771	(under) 0.2039
$t1_{19}$	100	0.014	10x	26	0.3993	(OVER) 0.3853
$t1_{20}$	98	-0.031	10x	70	0.0803	(OVER) 0.1113
$t1_{22}$	90	0.261	1000id	222	0.3272	(OVER) 0.0662
$t1_{25}$	79	2.299	1000id	56	1.4156	(under) 0.8834
$t1_{26}$	76	0.111	1000id	203	0.3824	(OVER) 0.2714
$t1_{27}$	52	-0.178	100id	190	0.0118	(OVER) 0.1898
$t1_{28}$	47	0.075	100id	192	0.0868	(OVER) 0.0118
$t1_{30}$	41	0.031	100id	39	0.3448	(OVER) 0.3138
$t1_{31}$	38	0.153	100id	208	0.0784	(under) 0.0746
$t1_{37}$	0	0.112	10id	49	0.0556	(under) 0.0564
$t1_{21}$	95	-0.065	10x	281	-0.0077	(OVER) 0.0573
$t1_{35}$	14	-0.004	10id	205	0.0262	(OVER) 0.0302

Derivations: T_2

$$\rho_{T_1+T_2} = \begin{pmatrix} 1 + (|a|^2 - 1)\epsilon_1 & ab^*\epsilon_2 \\ a^*b\epsilon_2 & |b|^2\epsilon_1 \end{pmatrix} \because \epsilon_1 = e^{-t/T_1}, \epsilon_2 = e^{-t/T_2} \tag{6}$$

Multiplying by the H gate matrix, we get probabilities (no SPAM)

$$p_0 = 1 + (|a|^2 - 1)\epsilon_1 + ab^*\epsilon_2 \tag{7}$$

$$p_1 = a^*b\epsilon_2 + |b|^2\epsilon_1 \tag{8}$$

T_2

$$c = -\frac{1}{2}\log\left(\frac{b(1-p_{0\to 1})+p_{1\to 0}}{b(p_{0\to 1})+(1-p_{1\to 0})}\right), \text{ where } b = \frac{1-\epsilon_2}{1+\epsilon_2} \quad (9)$$

- Formula modifies to above; **Assuming:** we measure in H basis
- We ran left half with extra measurements before circuit and right half without to check for correlated errors

Results: T_2

- Let $\Delta_\epsilon c = |c_{\text{pred}} - c_{\text{real}}|$
- Pre-measurements have obvious benefits in experiments, indicating correlated errors are lower
  - 1000id $\Delta_\epsilon c$ with measures: 0.83, 0.25, 0.68
  - 1000id $\Delta_\epsilon c$ without measures: 1.82, 3.37
  - 100x $\Delta_\epsilon c$ with measures: 1.75, 1.38, 0.38
  - 100x $\Delta_\epsilon c$ without measures: 3.02, 3.18
- Our formula is erroneous with mean $\Delta_\epsilon c = 2.44$ and $\sigma = 1.53$
- We have **overestimated** error in almost every case by orders of magnitude.
- There is no difference between single, double, and triple connected qubits in T_2 error

Accounting for IBM's Correction

- Higher depth errors (1000x) are lower than short depth (10x) errors: **Hypothesis:** What is net 'speed' for T_2 is lower?
  - We know it is an exponential decay, but what k is if $e^{(-t/T_2)^k}$
  - What if the IBM is doing short depth correction
- We change interpolation 'speed' from $t/T_2 \rightarrow (t/T_2)^{0.1}$ i.e $\epsilon_2 \rightarrow \epsilon_2^{0.1}$ (binary search gave 0.12)
- With these errors we now recalculate our errors and find that the mean $\Delta_\epsilon c = 0.95$ and $\sigma = 0.67$. (from mean $\Delta_\epsilon c = 2.44$ and $\sigma = 1.53$)

Results of Pre-measurement

Results before diagram: We measure the expectation of measurement (no statistics, just expectation)

- Everything under 2% is just readout error
- Isolated qubits have lower errors
- Bad qubits have higher errors
- Difference between double, and triple connected qubits is not evident
- No other patterns are visible
- Pre-measurement is objectively better for a lower variance (measurement collapses entanglement)

Results

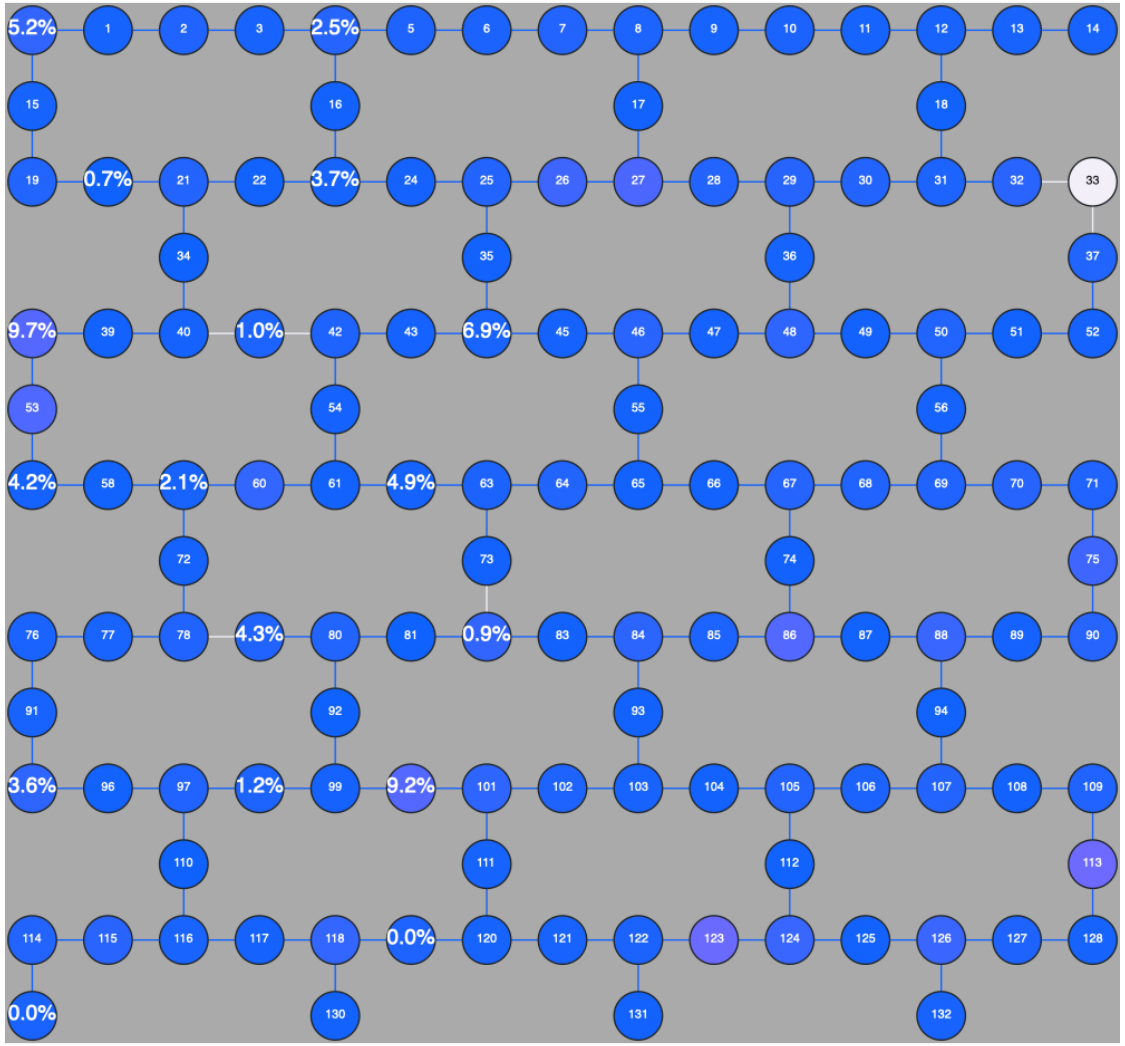


Figure: T2 results

Next Steps

- Add bounds
- Run on H_2 Hartree Fock Circuit
- Use $\Delta_\epsilon c$ as a metric to benchmark noise models